

Graphs, permutations and topological groups

Rögnvaldur G. Möller

Abstract. This is to explain how to prepare a contribution for publication in *Totally disconnected groups, graphs and geometry*.

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Introduction

The aim of this paper is to discuss various links between permutation groups, graphs and topological groups. The action of a group on a set can be used to define a topology on the group called the *permutation topology*. The earliest references for this topology are the paper [45] by Maurer and the paper [37] by Karrass and Solitar. This topology opens up the possibility of applying concepts and results from the theory of topological groups in permutation group theory. One can also go the otherway and apply ideas from permutation group theory to problems about topological groups. In particular some simple constructions of graphs that are commonly used in permutation group theory can be applied.

In the first section we discuss the language we will be using when working with graphs and permutation groups.

In the Section 2 we look at the definition of the permutation topology and consider applications of the theory of topological groups to questions about permutation groups. The main result is a theorem of Schlichting from [60]. This is a theorem about permutation groups but Schlichtings proof uses notions from functional analysis and result of Iwasawa [36] about topological groups.

In the third and fourth sections we discuss applications of techniques and ideas from permutation groups theory and graph theory to the theory of topological groups.

In Section 3 Willis's structure theory of totally disconnected locally compact groups is in the limelight. Willis's paper [71] helped spark a new interest in totally disconnected locally compact groups. Later work by Willis and others has shown that the concepts of the theory have many applications and are open to various interpretations. Most of the material in this section comes from the paper [49] where the basics of Willis's theory are given a graph theoretic interpretation.

The fourth and last section discuss an analogue of a Cayley graph, called a rough Cayley graph, that one can construct for a compactly generated totally disconnected group. The definition of this graphs is in Section 4.1. In that section it is also shown that this graph is a quasi-isometry invariant of the group. In the

latter parts of Section 4 it is shown how one uses rough Cayley graphs by developing an analogue for compactly generated totally disconnected locally compact groups of the theory of ends of groups and the theory of groups with polynomial growth.

There are various other topics that should be discussed in survey like this. The study of random walks on groups and graphs is another place where graphs, permutations and topological groups meet. The book by Woess [82] is an excellent introduction to this field. Another meeting place for graphs, permutations and topological groups is the theory of groups acting on trees. In particular one could mention the theory of harmonic analysis and representation theory of groups acting on trees, see the book by Figa-Talamanca and Nebbia[27] and the theory of tree lattices, see the book by Bass and Lubotzky [4]. Then there is the topic of generic elements and subgroups, see the papers by Bhattacharjee [7] and [8] and the paper by Abert and Glasner [2]. And then I have not even mentioned the manifold appearance of our trio of graphs, permutations and topology in model theory. Describing all these topics would have meant a book length paper, so I limited myself to discussing those aspects which I think I know.

1. Languages

1.1. A language for graphs.

We will both be discussing undirected graphs (usually called *graphs*) and directed graphs (usually called *digraphs*).

Our undirected graphs are without loops and multiple edges. Thus one can think of a (undirected) graph Γ as an ordered pair $(V\Gamma, E\Gamma)$ where $V\Gamma$ is a set and $E\Gamma$ is a set of two element subsets of $V\Gamma$. The elements of $V\Gamma$ are called *vertices* and the elements of $E\Gamma$ are called *edges*.

Vertices α and β in a graph Γ are said to be *neighbours*, or *adjacent*, if $\{\alpha, \beta\}$ is an edge in Γ . The *valency* of a vertex is the number of neighbours. If every vertex in Γ has finite valency then we say that the graph is *locally finite*. A *walk* of length n from α to β is a sequence $\alpha = \alpha_0, \alpha_1, \dots, \alpha_n = \beta$ of vertices, such that α_i and α_{i+1} are adjacent for $i = 0, 1, \dots, n-1$. If all the vertices in a walk are distinct then it is called a *path*. A *ray* in a graph is a sequence $\alpha_0, \alpha_1, \dots$ of distinct vertices such that α_i is adjacent to α_{i+1} for all i . A graph is connected if for any two vertices α and β there is a walk from α to β . Let $d(\alpha, \beta)$ denote the length of a shortest walk from a vertex α to a vertex β . If Γ is connected then d is a metric on $V\Gamma$. Let A be a set of vertices in Γ . The subgraph of Γ *spanned by* A is the graph having A as a vertex set and the edge set is the set of all edges in Γ whose end vertices are both in A . We say that A is *connected* if the subgraph spanned by A is connected. The *connected components* (or just *components*) of a graph are the maximal connected sets of vertices.

Formally we define a *digraph* Γ to be a pair $(V\Gamma, E\Gamma)$ where $V\Gamma$ is a set and $E\Gamma$ is a set of ordered pairs of distinct elements of $V\Gamma$, i.e. $E\Gamma \subseteq V\Gamma \times V\Gamma \setminus \{(\alpha, \alpha) \mid \alpha \in V\Gamma\}$. An edge (α, β) is thought of as an “arrow” starting in α and ending in β . We define the *in-valency* of a vertex β in Γ as the number of edges of the type

(α, β) (number of edges going “in to” β). Similarly, the *out-valency* of a vertex α is the number of edges (α, β) that go “out of” α . A digraph is said to be *locally finite* if every vertex has finite in- and out-valencies.

On the basis of a directed graph Γ one can define a undirected graph $\bar{\Gamma}$ with the same vertex set and two vertices α and β adjacent if and only if (α, β) or (β, α) is an edge in the directed graph Γ .

A *walk* in a digraph Γ is a sequence $\alpha_0, \dots, \alpha_n$ such that (α_i, α_{i+1}) or (α_{i+1}, α_i) is an edge in Γ for $i = 0, \dots, n-1$. (That is to say, $\alpha_0, \dots, \alpha_n$ is a walk in the undirected graph $\bar{\Gamma}$.) An *arc*, more specifically an *n-arc*, is a sequence $\alpha_0, \dots, \alpha_n$ of distinct vertices such that (α_i, α_{i+1}) is an edge in Γ for $i = 0, \dots, n-1$. The set of *descendants* of a vertex α is defined as the set

$$\text{desc}(\alpha) = \{\beta \in VT \mid \text{there exists an arc from } \alpha \text{ to } \beta\}.$$

The set $\text{desc}_k(\alpha)$ is defined as the set of all vertices β such that the shortest arc from α to β has length k . The set of *ancestors* of a vertex α , denoted $\text{anc}(\alpha)$, is defined as the set of all vertices β such that $\alpha \in \text{desc}(\beta)$, i.e. $\beta \in \text{anc}(\alpha)$ if and only if there exists an arc from β to α .

Finally, we review the definition of a *Cayley graph* of a group. Let G be a group and S a subset of G . The (undirected) Cayley graph $\text{Cay}(G, S)$ of G with respect to S has G as the vertex set and $\{g, h\}$ is an edge if $h = gs$ or $h = gs^{-1}$ for some s in S . The Cayley graph $\text{Cay}(G, S)$ is connected if and only if S generates G . The left regular action of G on itself gives us a transitive action of G as a group of graph automorphisms on $\text{Cay}(G, S)$.

It is perhaps more natural to think of the Cayley graph as a directed graph where the vertex set is G and (g, h) is an edge if and only if $h = gs$ for some $s \in S$. In these notes, however, it is convenient to think of Cayley graphs as being undirected.

1.2. A language for permutation groups.

Let G be a group acting on a set Ω . The image of a point $\alpha \in \Omega$ under an element $g \in G$ is written as $g\alpha$. The group of all permutations of a set Ω is called the *symmetric group on Ω* and denoted with $\text{Sym}(\Omega)$.

The action of G on Ω induces a natural homomorphism $G \rightarrow \text{Sym}(\Omega)$. If this homomorphism $G \rightarrow \text{Sym}(\Omega)$ is injective then we say that G acts *faithfully* on Ω . A group G acting faithfully on a set Ω can be regarded as a subgroup of $\text{Sym}(\Omega)$, and we say that G is a permutation group on Ω .

The action is said to be *transitive* if for every two points $\alpha, \beta \in \Omega$ there is some element $g \in G$ such that $g\alpha = \beta$. For a point $\alpha \in \Omega$, the subgroup

$$G_\alpha = \{g \in G \mid g\alpha = \alpha\}$$

is called the *stabilizer* in G of the point α . If Δ is a subset of Ω then the *pointwise stabilizer* $G_{(\Delta)}$ of Δ is defined as the subgroup of all the elements in G that fix every element of Δ , that is,

$$G_{(\Delta)} = \{g \in G \mid g\delta = \delta \text{ for every } \delta \in \Delta\} = \bigcap_{\delta \in \Delta} G_\delta.$$

The *setwise stabilizer* $G_{\{\Delta\}}$ of Δ is defined as the subgroup consisting of all elements of G that leave Δ invariant, that is,

$$G_{\{\Delta\}} = \{g \in G \mid g\Delta = \Delta\}.$$

The orbit (for emphasis sometimes called a G -orbit) of a point α is the set $\{g\alpha \mid g \in G\}$.

Suppose U is a subgroup of a group G . The group G acts on the set G/U of right cosets of U and this action is transitive. The image of a coset hU under an element $g \in G$ is $(gh)U$. Conversely, if G acts transitively on some set Ω and α is a point in Ω then Ω can be identified with G/G_α . Here “identified” means that there is a bijective map $\theta : \Omega \rightarrow G/G_\alpha$ such that for every $\omega \in \Omega$ and every element $g \in G$ we get $\theta(g\omega) = g\theta(\omega)$.

The orbits of stabilizers of points in Ω are called *suborbits*, that is, the suborbits are sets of the form $G_\alpha\beta$ where $\alpha, \beta \in \Omega$. Orbits of G on the set of ordered pairs of elements from Ω are called *orbitals*. When G is transitive on Ω one can, for a fixed point $\alpha \in \Omega$, identify the suborbits of G_α with the orbitals: the suborbit $G_\alpha\beta$ is identified with the orbital $G(\alpha, \beta)$. The number of elements in a suborbit $G_\alpha\beta$, often called the *length* of the suborbit, is given by the index $|G_\alpha : G_\alpha \cap G_\beta|$. In general the number of elements in the orbit $U\alpha$ of a subgroup U is equal to the index $|U : U \cap G_\alpha|$.

Define a (*directed*) *orbital graph* as a graph with vertex set Ω and the edge set is a union of orbitals. The action of G on its vertex set Ω induces an action of G as a group of automorphisms on the graph Γ , because if (α, β) is an edge in Γ then $(g\alpha, g\beta)$ is in the same orbital and therefore also an edge in Γ . One can also define undirected orbital graphs. Then we take Ω as the vertex set and the edge set is a union of sets of the form $G\{\alpha, \beta\}$ (the G -orbit of the undirected pair $\{\alpha, \beta\}$) where α and β are distinct points in Ω .

A *block of imprimitivity* for G is a subset Δ of Ω such that for every $g \in G$, either $g\Delta = \Delta$ or $\Delta \cap (g\Delta) = \emptyset$. The existence of a non-trivial proper block of imprimitivity Δ (*non-trivial* means that $|\Delta| > 1$ and *proper* means that $\Delta \neq \Omega$) is equivalent to the existence of a non-trivial proper G -invariant equivalence relation $\sim$ on Ω . When G acts transitively on Ω , the block Δ and its translates under G give the $\sim$ -classes, and conversely if $\sim$ is a non-trivial proper G -invariant equivalence relation then the $\sim$ -classes are non-trivial proper blocks of imprimitivity for G . When $\sim$ is a G -invariant equivalence relation on Ω then G permutes the $\sim$ -classes and thus G acts on the set $\Omega/\sim$ of equivalence classes. If there is no non-trivial proper G -invariant equivalence relation on Ω we say that G acts primitively on Ω . In most (all?) books on permutation groups it is shown that, if G acts transitively on Ω then G acts primitively on Ω if and only if G_α is a maximal subgroup of G for every $\alpha \in \Omega$. Part of the proof of this is to show that if $G_\alpha < H < G$ then $H\alpha$ is a non-trivial proper block of imprimitivity. A further useful property is that if N is a normal subgroup of G then the orbits of N on Ω are blocks of imprimitivity for G .

Recent books covering this material (and of-course lots more) are [9], [12] and [18].

2. The permutation topology

2.1. Definition of the permutation topology.

Let G be a group acting on a set Ω . The action can be used to introduce a topology on G . The topology of a topological group is completely determined by a neighbourhood basis of the identity element. The *permutation topology* on G is defined by choosing as a neighbourhood basis of the identity the family of pointwise stabilizers of finite subsets of Ω , i.e. a neighbourhood basis of the identity is given by the family of subgroups

$$\{G_{(\Phi)} \mid \Phi \text{ is a finite subset of } \Omega\}.$$

There are other ways to define the permutation topology. Think of Ω as having the discrete topology and elements of G as maps $\Omega \rightarrow \Omega$. Then the permutation topology is equal to the topology of pointwise convergence, and it is also the same as the compact-open topology. A sequence (g_i) of elements in G has an element $g \in G$ as a limit if and only if for every $\alpha \in \Omega$ there is a number N (depending on α) such that $g_n\alpha = g\alpha$ for every $n \geq N$.

For the case of a permutation group G acting on a countable set Ω we can also define a metric on G inducing the permutation topology. The basic idea is that two permutations g and h are “close” to each other if they agree on “many” points. Here is one way of defining a metric d on G that induces the permutation topology:

Enumerate the points in Ω as $\alpha_1, \alpha_2, \dots$. Take two elements $g, h \in G$. Let n be the smallest number such that $g\alpha_n \neq h\alpha_n$ or $g^{-1}\alpha_n \neq h^{-1}\alpha_n$. Set $d(g, h) = 1/2^n$. Then d is a metric on G and induces the permutation topology.

From the definition of the permutation topology we can immediately characterize open subgroups in G :

A subgroup of G is open if and only if it contains the pointwise stabilizer of some finite set of points.

Various properties of the action of G on Ω are reflected in properties of this topology on G .

The permutation topology on G is Hausdorff if and only if the action of G on Ω is faithful. Moreover, G is totally disconnected if and only if the action is faithful.

We say that G is a *closed permutation group* if it is a closed subgroup of $\text{Sym}(\Omega)$, where $\text{Sym}(\Omega)$ has the permutation topology. Closed permutation groups can be characterized in the following way, see [11, Section 2.4].

Proposition 2.1. *A permutation group G on a set Ω is closed if and only if G is the full automorphism group of some first order structure on Ω .*

A first order structure on Ω is a collection of:

- constants that belong to Ω ,
- functions defined on Ω and taking their values in Ω ,

- relations defined on Ω .

It is easy to show that the automorphism group of such a structure is closed. For a group G acting on Ω we define a relational structure on Ω such that for $n = 1, 2, \dots$ we get one n -ary relation for each orbit of G on n -tuples of Ω . This structure is called the *canonical relational structure* for the action of G on Ω . If G is a closed permutation group then G is the full automorphism group of this structure.

The group actions we are interested in are not necessarily faithful. We say that G is closed in the permutation topology if the image of G in $\text{Sym}(\Omega)$ under the natural homomorphism is closed. This condition is equivalent to the condition that stabilizers of points are closed subgroups of G .

Compactness has a very natural interpretation in the permutation topology. A subset of a topological space is said to be *relatively compact* if it has compact closure. The following lemma was formulated by Woess for automorphism groups of locally finite connected graphs.

Lemma 2.2. ([81, Lemma 1 and Lemma 2]) *Let G be a permutation group acting transitively on a countable set Ω . Assume that G is closed and that all suborbits are finite.*

- (i) *The stabilizer G_α of a point $\alpha \in \Omega$ is compact.*
- (ii) *A subset A of G is relatively compact in G if and only if the set $A\alpha$ is finite for every α in Ω .*

Furthermore, if A is a subset of G and $A\alpha$ is finite for some $\alpha \in \Omega$ then $A\alpha$ is finite for all α in Ω .

Proof. (i) Because G is metrizable we only need to show that each sequence (g_n) of elements in G_α contains a convergent subsequence g_{n_i} .

Enumerate the elements of Ω as $\alpha_1, \alpha_2, \dots$. Since the orbit of α_1 under G_α is finite we can find a number n_1 such that $g_n\alpha_1 = g_{n_1}\alpha_1$ for infinitely many values of n . Then we choose $n_2 > n_1$ such that $g_{n_2}\alpha_1 = g_{n_1}\alpha_1$ and there are infinitely many values of n such that $g_n\alpha_2 = g_{n_2}\alpha_2$. We continue in this way and define a subsequence (g_{n_i}) which is the convergent subsequence we are looking for.

(ii) Suppose that A^- , the closure of A in G , is compact. Then for a point $\alpha \in \Omega$ there is a finite open covering of A^- by sets of the type gG_α ; that is to say, we can find $g_1, \dots, g_n \in \text{Aut}(\Gamma)$ such that $A^- \subseteq \bigcup_{i=1}^n g_i G_\alpha$. Then $A\alpha \subseteq \{g_1\alpha, \dots, g_n\alpha\}$.

Now suppose that $A\alpha = \{\alpha_1, \dots, \alpha_n\}$. Let g_i be an element in A such that $g_i\alpha = \alpha_i$. Then $A \subseteq \bigcup_{i=1}^n g_i G_\alpha$. The latter set is compact, so the closure of A is compact. $\square$

This Lemma implies that if G is a closed transitive permutation group on a countable set Ω such that all suborbits are finite then G , with the permutation topology, is a locally compact totally disconnected group. In particular the automorphism group of a locally finite transitive graph is a locally compact totally disconnected group.

A subgroup H in a topological group G is said to be cocompact if G/H is a compact space. This concept has also a natural interpretation when looking at the permutation topology.

Lemma 2.3. ([54, Proposition 1], cf. [49, Lemma 7.5]) *Let G be a group acting transitively on a set Ω and assume that G is closed in the permutation topology and all suborbits are finite. Then a subgroup H of G is cocompact if and only if H has only finitely many orbits on Ω .*

Proof. Suppose first that H is cocompact. This means that both the spaces of right and left cosets of H in G are compact. Let X denote the set of right cosets of H in G . The quotient map $\pi : G \rightarrow X$ is open. The family of cosets $\{gG_\alpha\}_{g \in G}$ is an open covering of G and hence $\{\pi(gG_\alpha)\}_{g \in G}$ is an open covering of X . We are assuming that X is compact, so there is a finite subcovering $\pi(g_1G_\alpha), \dots, \pi(g_nG_\alpha)$ of X . Then $G = Hg_1G_\alpha \cup \dots \cup Hg_nG_\alpha$ and therefore $\Omega = H(g_1\alpha) \cup \dots \cup H(g_n\alpha)$.

Suppose now that H has only finitely many orbits on Ω , say there are elements $g_1, \dots, g_n$ such that $\Omega = H(g_1\alpha) \cup \dots \cup H(g_n\alpha)$. Then $G = Hg_1G_\alpha \cup \dots \cup Hg_nG_\alpha$ and $X = \pi(g_1G_\alpha) \cup \dots \cup \pi(g_nG_\alpha)$. Each of the sets $\pi(g_iG_\alpha)$ is compact. Hence X , the set of right cosets of H in G , is compact, because it is a union of finitely many compact sets. $\square$

We could also start with a topological group G . For an open subgroup U we set $\Omega = G/U$, the space of left cosets. The stabilizers in G of points in Ω are conjugates of U and thus also open subgroups in G . The stabilizer of a finite set $\Phi = \{\alpha_1, \dots, \alpha_n\}$ of points is just the intersection of the open subgroups $G_{\alpha_1}, \dots, G_{\alpha_n}$ and is thus open in G . From this we see that the permutation topology coming from the action of G on Ω is contained in the topology on G . If the topological group G is assumed to be a totally disconnected and locally compact, then we can choose U to be a compact open subgroup of G (by a theorem of Van Dantzig [14]). In particular, the stabilizers in G of points in Ω are all compact (they are conjugates of U). The second part of the Lemma above also holds for the action of G on Ω , so a subset A of G has compact closure if and only if $A\alpha$ is finite for every $\alpha \in \Omega$. This implies that $|G_{\alpha\beta}| \leq \infty$ for all points α and β in Ω . The reason is that $|G_{\alpha\beta}| = |G_\alpha : G_\alpha \cap G_\beta|$ and this index is finite because $G_\alpha \cap G_\beta$ is an open subgroup of the compact group G_α .

2.2. Suborbits and the modular function. In this section the general assumption will be that G is a closed permutation group acting on a set Ω and that all suborbits of the group G are finite.

Under the above assumptions the group G is a locally compact totally disconnected group. In this section we want to interpret the modular function on G in relation to the action of G on Ω . The connection between suborbits and the modular function can be seen from the following argument due to Schlichting [59], see also [68].

Let μ be a right invariant Haar-measure on G . Define the modular function Δ so that if A is a measurable set then $\mu(gA) = \Delta(g)\mu(A)$.

Lemma 2.4. ([59, Lemma 1], cf. [68, Theorem e]) *Let G be a closed permutation group acting transitively on a set Ω and assume that all suborbits of G are finite. Let Δ denote the modular function on G . If h is an element in G with $h\alpha = \beta$ then*

$$\Delta(h) = \frac{|G_\beta\alpha|}{|G_\alpha\beta|} = \frac{|G_\alpha : G_\alpha \cap h^{-1}G_\alpha h|}{|G_\alpha : G_\alpha \cap hG_\alpha h^{-1}|}.$$

Proof. Then, with μ denoting the right invariant Haar measure on G ,

$$\begin{aligned} |G_\beta\alpha| &= |G_\beta : G_{\alpha\beta}| \\ &= \mu(G_\beta)/\mu(G_{\alpha\beta}) \\ &= \mu(hG_\alpha h^{-1})/\mu(G_{\alpha\beta}) \\ &= \Delta(h)\mu(G_\alpha)/\mu(G_{\alpha\beta}) \\ &= \Delta(h) |G_\alpha : G_{\alpha\beta}| \\ &= \Delta(h) |G_\alpha\beta|. \end{aligned}$$

And we see that $\Delta(h) = |G_\beta\alpha|/|G_\alpha\beta|$. □

For an orbital $A = G(\alpha, \beta)$ we define the *paired orbital* as the orbital $A^* = G(\beta, \alpha)$. The pairing of orbitals also gives us a pairing of suborbits, where the suborbit $G_\alpha\beta$ is paired to a suborbit $G_\alpha\gamma$ where γ is a point in Ω such that (α, γ) is in $G(\beta, \alpha)$. Considering the directed orbital graph for the orbital $G(\alpha, \beta)$. The size of the suborbit $G_\alpha\beta$ is the out-valency of this graph and the size of the paired suborbit $G_\alpha\gamma$ is the in-valency of this graph. The above Lemma implies.

Proposition 2.5. ([68, Theorem 1]) *Let G be a closed permutation group acting transitively on a set Ω . Assume that all suborbits of G are finite. Then the lengths of paired suborbits are equal if and only if G is unimodular.*

One property of the modular function that we have not yet used is that it is a homomorphism.

Corollary 2.6. *Let G be a closed permutation group acting transitively on a set Ω . Let $K = \ker(\Delta)$. Then*

- (a) *If $\alpha \in \Omega$ then $G_\alpha \leq K$.*
- (b) *Let G' be the derived group of G . Then $G' \leq K$.*
- (c) *If g is an torsion element of G then $g \in K$.*

Proof. Follows directly from the definitions and the fact that G/K is isomorphic to a multiplicative subgroup of the positive real numbers. □

Corollary 2.7. *Let G be a closed permutation group acting transitively on a set Ω . Assume that all suborbits of G are finite. If G is also primitive then paired suborbits have equal length.*

Proof. Let $K = \ker(\Delta)$. Then, because G is primitive and K is normal in G , we know that either $K = \{e\}$ or K is transitive. If $K = \{e\}$ then G is abelian and must in fact be trivial. If K is transitive then the result of Lemma 1 implies that lengths of paired suborbits are equal. $\square$

For a long time it was an open question, posed by Peter M. Neumann, whether one could have a group acting primitively on a set Ω with a finite suborbits paired to an infinite one. Such examples were constructed by David M. Evans [26].

Consider a connected graph Γ and assume that G is a closed subgroup of $\text{Aut}(\Gamma)$ that acts transitively on Γ and that all suborbits are finite. For convenience we think of each undirected edge $\{\alpha, \beta\}$ in Γ as consisting of two directed edges (α, β) and (β, α) . Each directed edge $e = (\alpha, \beta)$ will be labeled by a number

$$\Delta_e = \frac{|G_{\beta}\alpha|}{|G_{\alpha}\beta|}.$$

Observe that $\Delta_{(\alpha, \beta)} = \Delta_{(\beta, \alpha)}^{-1}$. Furthermore, take note that if g is an element of G such that $g\alpha = \beta$ and $e = (\alpha, \beta)$ then $\Delta(g) = \Delta_e$. Suppose g is an element of G and $g\alpha = \gamma$ and that there is a vertex β in X such that (α, β) and (β, γ) are edges in X . Find elements g_1 and g_2 in G such that $g_1\alpha = \beta$ and $g_2\beta = \gamma$. Then $g\alpha = g_2g_1\beta$ and from the formula above for the modular function we can deduce that $\Delta(g) = \Delta(g_2g_1)$. We also find that

$$\Delta(g) = \Delta(g_2g_1) = \Delta(g_2)\Delta(g_1) = \Delta_{(\beta, \gamma)}\Delta_{(\alpha, \beta)}.$$

This can be extended to directed walks of arbitrary length, so that if $g\alpha = \beta$ then we take a directed walk from α to β , enumerate the edges in the walk as $e_1, \dots, e_k$ and then

$$\Delta(g) = \Delta_{e_1} \cdots \Delta_{e_k}.$$

Hence the labeled graph completely describes the modular function on G . This idea can be found in the paper by Bass and Kulkarni [3].

Suppose now not only that all the suborbits of G are finite but also that there is a finite upper bound m on their length. Looking again at Lemma 1 we see that then the image of the modular function is bounded. The image of the modular function is a subgroup of the multiplicative group of positive rational numbers and the only bounded subgroup is the trivial subgroup. Hence the image of Δ is trivial and thus G is unimodular. We have now deduced the following result of Praeger (unpublished).

Corollary 2.8. *Let G be a closed permutation group acting transitively on a set Ω such that all suborbits of G are finite. If there is a finite bound on the length of suborbits then paired suborbits have equal length.*

The next result, also due to Praeger [57], uses the modular function to find information about graph structure.

Theorem 2.9. ([57]) *Let Γ be an infinite connected, vertex and edge transitive directed graph with finite but unequal in- and out-valence. Then there is a graph epimorphism φ from Γ to the directed integer graph $\mathbf{Z}$. For each $i \in \mathbf{Z}$, the inverse image $\varphi^{-1}(i)$ is infinite.*

Proof. Write $G = \text{Aut}(\Gamma)$. Let $q = d^-/d^+$ where d^+ is the out-valence of Γ and d^- is the in-valence of Γ . Consider an edge (α, β) in Γ . Because G acts transitively on the edges of Γ we can conclude that $|G_\alpha\beta| = d^+$ and $|G_\beta\alpha| = d^-$ and hence if $g\alpha = \beta$ then $\Delta(g) = d^-/d^+$. Using the graph Γ to calculate the modular function in a similar way as described above we conclude that for every element $g \in G$ there is a number i such that $\Delta(g) = q^i$. Hence, if K denotes the kernel of the modular function then $G/K = \mathbf{Z}$. Fix a vertex α_0 in Γ and define a map $\varphi : V\Gamma \rightarrow \mathbf{Z}$ so that $\varphi(\beta) = i$ if there is an element g in G such that $g\alpha_0 = \beta$ and $\Delta(g) = q^i$. It is clear that the choice of g is immaterial. From the way one uses Γ to calculate the modular function we see that if (α, β) is an edge in Γ then $\varphi(\beta) = \varphi(\alpha) + 1$ which implies that φ is a homomorphism from Γ to the directed integer graph (the vertex set is the set of integers and the edge set is the set of all ordered pairs $(n, n+1)$). (A similar proof of Praeger's result is in a paper by Evans [25].) $\square$

Remark. In the next section *highly arc transitive* digraphs are discussed. The digraph Γ in the theorem above need not be highly arc transitive, but it is easy to show that if d^+ and d^- are coprime then Γ must be highly arc transitive.

The next result we discuss is a remarkable theorem of Schlichting [60].

Theorem 2.10. ([60]) *Let G be a group acting transitively on a set Ω . Then there is a finite bound on the sizes of suborbits of G if and only if there is a G -invariant equivalence relation $\sim$ on Ω with finite classes such that the stabilizers of points in the action of G on $\Omega/\sim$ are finite.*

This is a theorem about permutation groups but Schlichting's proof utilizes various concepts from functional analysis and a theorem of Iwasawa [36, Theorem 1]. Later this theorem was rediscovered by Bergman and Lenstra [10], who gave a group theoretical/combinatorial proof.

The theorem of Iwasawa that Schlichting uses in his proof is about the relationship between the properties $[IN]$ and $[SIM]$ for a topological group.

Definition 2.11. A locally compact group is said to have property $[IN]$ if there is a compact neighbourhood K of the identity (i.e. K contains an open set containing the identity) that is invariant under conjugation by elements in G .

A locally compact group is said to have property $[SIN]$ if every neighbourhood of the identity contains a compact neighbourhood K of the identity that is invariant under conjugation by elements in G .

Let us start by relating these two properties to the permutation topology.

Proposition 2.12. *Let G be a permutation group acting transitively on a set Ω and assume that all suborbits are finite. Then G with the permutation topology has property $[IN]$ if and only if there is a finite bound on the sizes of suborbits.*

Proof. First assume that G has property $[IN]$. Suppose $\alpha, \beta \in \Omega$. We want to find a constant upper bound, independent of α and β , for the size of the suborbit $G_\alpha\beta$. Let K be a compact neighbourhood of the identity that is invariant under conjugation. Since K contains an open neighbourhood of the identity we can find a finite set Φ such that $G_{(\Phi)} \subseteq K$. Take a point γ in Ω . Because K is compact then $|K\gamma| = m < \infty$. Let k be a number such that $|G_\delta : G_{(\Phi)}| < k$ for all points $\delta \in \Phi$. Find an element $f \in G$ such that $f\beta = \gamma$ and set $\alpha' = f\alpha$. Then clearly $|G_\alpha\beta| = |G_{\alpha'}\gamma|$. Then we find an element h such that $\alpha' \in h\Phi$. Note that $G_{(h\Phi)} = hG_{(\Phi)}h^{-1}$. Since K is invariant under conjugation and $G_{(\Phi)} \subseteq K$ we conclude that $G_{(h\Phi)} \subseteq K$. Therefore $|G_{(h\Phi)}\gamma| \leq |K\alpha| = m$. We also know that $|G_{\alpha'} : G_{(h\Phi)}| \leq k$ and thus $|G_{\alpha'}\gamma| \leq k|G_{(h\Phi)}\gamma| \leq km$. Hence we have a constant upper bound km on the size of all suborbits of G .

Now assume that there is a finite bound m on the sizes of suborbits. For a finite subset Φ of Ω we let $m(\Phi)$ denote the size of the largest suborbit of $G_{(\Phi)}$. We choose a finite subset Φ such that $m_0 = m(\Phi)$ is as small as possible. Define $A = \bigcup_{g \in G} gG_{(\Phi)}g^{-1}$. The set A is clearly invariant under conjugation by elements of G and clearly A is open. We want to show that A has compact closure. In order to do so we only need to show that $A\alpha$ is finite for some $\alpha \in \Omega$. Choose $\alpha \in \Omega$ such that $|G_{(\Phi)}\alpha| = m(\Phi)$. Suppose $|A\alpha| \geq n > m(\Phi)$. Take elements $f_1, \dots, f_n \in A$ such that the elements $f_1\alpha, \dots, f_n\alpha$ are all distinct. Since $f_1, \dots, f_n \in A = \bigcup_{g \in G} gG_{(\Phi)}g^{-1}$, we can find elements $g_1, \dots, g_n \in G$ such that $f_i \in g_iG_{(\Phi)}g_i^{-1} = G_{(g_i\Phi)}$. Write $\Phi_i = g_i\Phi$ and set $E = \{f_1\alpha, \dots, f_n\alpha\} \cup \Phi_1 \cup \dots \cup \Phi_n$. Then $m(E) = m_0$. Choose an element $\delta \in \Delta$. There is for each $i = 1, \dots, n$ an element $h_i \in G_{(E)}$ such that $h_i\delta = f_i\delta$ and therefore $h_i^{-1}f_i \in G_\delta$. But $h_i^{-1}f_i\alpha = f_i\alpha$ and we can conclude that $|G_\delta\alpha| \geq n > m$ contrary to assumptions. (The above argument is related to the proof of Theorem 2.10 by Bergmann and Lenstra [10], but this version is from a lecture course given by Peter M. Neumann in Oxford 1988–1989.) $\square$

Proposition 2.13. *Let G be a permutation group acting transitively on a set Ω and assume that all suborbits are finite. Then G with the permutation topology has property $[SIN]$ if and only if G is discrete (i.e. the stabilizers of points are finite).*

Proof. It is obvious that if G is discrete then G has property $[SIN]$.

Let us now assume that G has property $[SIN]$. Then, if α denotes a point in Ω , the open subgroup G_α contains a compact neighbourhood K of the identity

that is invariant under conjugation. Then for $g \in G$ we see that $K = gKg^{-1} \subseteq gG_\alpha g^{-1} = G_{g\alpha}$ and thus $K \subseteq \bigcap_{g \in G} gG_\alpha g^{-1} = \bigcap_{g \in G} G_{g\alpha}$. Because G is assumed to be transitive we conclude that K must fix every point of Ω . But we are also assuming that G acts faithfully on Ω so $K = \{e\}$ and G is discrete. $\square$

The theorem of Iwasawa mentioned above says that if a locally compact group has property $[IN]$ then there is a compact normal subgroup N such that G/N has property $[SIN]$.

Proof. (Theorem 2.10) Assume that there is a finite upper bound on the sizes of suborbits of G . By Proposition 2.12 the group G has property $[IN]$. Iwasawa's theorem gives us a compact normal subgroup N of G such that G/N has property $[SIN]$. The orbits of the normal subgroup N are the classes of a G -invariant equivalence relation $\sim$ on Ω and because N is compact these classes are all finite. The group $H = G/N$ certainly has property $[SIN]$, but it is not certain that H acts faithfully on $\Omega' = \Omega/\sim$ so we can not apply Proposition 2.13 directly. Note that if $\alpha \in \Omega'$ then H_α is an open subgroup of H . Since H has property $[SIN]$ there is a compact invariant neighbourhood K contained in H_α . As in the proof of Proposition 2.13 we conclude that K is contained in the kernel of the action of H on Ω' . Thus the kernel N' of the action of H on Ω' is an open subgroup of H . The group H/N' can be regarded as a permutation group on Ω' . From this we can conclude that the permutation topology on H/N' is discrete, which implies that the stabilizer in H/N' of a point $\alpha \in \Omega'$ must be finite.

The proof of the other direction is left to the reader. $\square$

Schlichtings theorem implies the following general result about totally disconnected locally compact groups.

Corollary 2.14. *Let G be a totally disconnected locally compact group.*

(i) *The group G has a compact open normal subgroup if and only if there is a compact open subgroup U and a number m_U such that $|U : U \cap gUg^{-1}| \leq m_U$ for all $g \in G$.*

(ii) *If there exists such a number m_U for one compact open subgroup U then there exists a number m_V for any compact open subgroup V such that $|V : V \cap gVg^{-1}| \leq m_V$ for all $g \in G$.*

Proof. (a) If G contains a compact open normal subgroup N then we can take $U = N$ and $m_N = 1$.

Assume now that U is a compact open subgroup and there is a number m_U such that $|U : U \cap gUg^{-1}| \leq m_U$. Set $\Omega = G/U$. Take a point α in Ω such that $U = G_\alpha$. Note that if $g \in G$ and $\beta = g\alpha$ then $|G_\alpha \beta| = |G_\alpha : G_\alpha \cap G_\beta| = |U : U \cap gUg^{-1}| \leq m_U$. Thus there is a finite upper bound on the sizes of suborbits and we can apply Schlichtings Theorem. Schlichtings Theorem provides us with an G -invariant equivalence relation $\sim$ on Ω with finite classes such that the stabilizer

of a $\sim$ -class acts like a finite group on $\Omega' = \Omega / \sim$. This in turn implies that N , the kernel of the action of G on Ω' , is a open normal subgroup of G . If $\alpha \in \Omega$ then $N\alpha$ is contained in the $\sim$ -class of α and thus $N\alpha$ is finite. Therefore N is also compact.

(b) From part (a) we get the existence of a compact open normal subgroup. Consider the action of G on the set $\Omega = G/V$. The orbits of N on Ω are all finite and give us the equivalence classes of G -invariant equivalence relation $\sim$. Let k be the number of element in a $\sim$ -class. The group G acts on $\Omega' = \Omega / \sim$ and the stabilizers of points in Ω' act like finite groups on Ω' . Thus the sizes of suborbits in the action of G on Ω' are bounded above by some number k . Let $\tilde{\alpha}$ denote the $\sim$ -class of an element $\alpha \in \Omega$. For elements α and β in Ω we see that $|G_\alpha\beta| \leq |G_{\tilde{\alpha}}\beta| \leq k|G_{\tilde{\alpha}}\beta| = kl$. From this it follows that $|V : V \cap gVg^{-1}| \leq kl$. $\square$

2.3. The theorems of Trofimov.

As mentioned above the automorphism group of a locally finite connected graph with the permutation topology is locally compact and totally disconnected. In this section we will discuss three theorems of Trofimov (see [66], [67], [69]). The conclusions in them all resemble the conclusion in Schlichtings Theorem, but none of the theorems is proved by referring to Schlichtings Theorem. We start with the earliest of these three theorems. First we must explain the terminology.

An automorphism g of a connected graph Γ (or an isometry of a metric space) is said to be *bounded* if there is a constant c such that $d_\Gamma(\alpha, g\alpha) \leq c$ for all vertices α in Γ .

Theorem 2.15. ([66]) *Let Γ be a locally finite transitive graph and $B(\Gamma)$ the subgroup of bounded automorphisms. The following assertions are equivalent:*

- (i) *The subgroup $B(\Gamma)$ is transitive.*
- (ii) *There is an equivalence relation $\sim$ with finite equivalence classes on the vertex set of Γ such that $B(\Gamma)$ acts on Γ like a free finitely generated abelian group.*

In a connected graph Γ (or a more general metric space) we define the ball of radius n with center in a vertex α as the set $B_n(\alpha) = \{\beta \in V\Gamma \mid d(\alpha, \beta) \leq n\}$. The bounded automorphisms are related to topological properties of $\text{Aut}(\Gamma)$ via the following result of Woess. With a later application in mind Woess's result is stated for metric spaces rather than just graphs. Recall that an element g in a topological group G is called an FC^- -element if the conjugacy class of g has compact closure.

Lemma 2.16. (Cf. [81, Lemma 4]) *Suppose G is a topological group acting transitively by isometries on a metric space X . Assume furthermore that the stabilizer in G of a point in X is a compact open subgroup and that for every value of n the ball $B_n(\alpha)$ is finite. An element $g \in G$ is bounded if and only if g is an FC^- -element of G .*

Proof. Suppose $g \in G$ acts as a bounded isometry on X . Find a number M such that $d(g\alpha, \alpha) \leq M$ for every $\alpha \in X$. For $h \in G$, write $g^h = hgh^{-1}$. It is clear that $d(g^h\alpha, \alpha) = d(gh^{-1}\alpha, h^{-1}\alpha) \leq M$ for every $\alpha \in V\Gamma$. Set $g^G = \{g^h \mid h \in G\}$.

We see that the set $g^G\alpha$ is finite and by Lemma 2.2 the conjugacy class g^G has compact closure.

Suppose now that the conjugacy class g^G has compact closure. Then, for every $\alpha \in X$ the set $g^G\alpha$ is finite. Take a number M such that $d(g^h\alpha, \alpha) \leq M$ for every $h \in G$. Take some $\beta \in X$. Choose $h \in G$ so that $\beta = h^{-1}\alpha$. Then $d(g\beta, \beta) = d(gh^{-1}\alpha, h^{-1}\alpha) = d(g^h\alpha, \alpha) \leq M$. So g acts on Γ as a bounded automorphism. $\square$

This connection with topological notions can be used to give a short proof of Theorem 2.15 where only elementary results from the theory of topological groups are used, see [48].

In a connected graph Γ we define the ball of radius n with center in a vertex α as the set $B_n(\alpha) = \{\beta \in V\Gamma \mid d(\alpha, \beta) \leq n\}$. A locally finite graph Γ is said to have *polynomial growth* if $|B_n(\alpha)|$ is bounded above by a polynomial in n . It is easy to see that this property does not depend on the choice of the vertex α . A finitely generated group G is said to have polynomial growth if its Cayley graph with respect to a finite generating set has polynomial growth (the choice of generating sets is immaterial, since having polynomial growth is a quasi-isometry invariant).

The second theorem of Trofimov related to Schlichtings Theorem is the following:

Theorem 2.17. ([67, Theorem 2]) *Suppose Γ is a connected locally finite graph with polynomial growth and G is a group that acts transitively on Γ . Then there is a G -invariant equivalence relation $\sim$ on the vertex set of Γ such that the equivalence classes of $\sim$ are finite such that if K denotes the kernel of the action of G on the equivalence classes then G/K is a finitely generated almost nilpotent group and the stabilizers in G/K of $\sim$ -classes are finite.*

It should be noted that Trofimov proves an even stronger result [67, Theorem 1], since he shows that it is possible to find an equivalence relation $\sim$ as described in Theorem 2.17 such that the stabilizer of a vertex in $\text{Aut}(\Gamma/\sim)$ is finite.

The theorem of Trofimov can be seen as a graph theoretical version of Gromov's celebrated theorem characterizing finitely generated groups with polynomial growth, see [30]. Indeed, Trofimov uses Gromov's theorem in his proof. A version of Gromov's theorem for topological groups has been proved by Losert in [41]. Woess in [81] used Losert's version of Gromov's theorem from [41] to give a short proof of Theorem 2.17.

We will be returning to polynomial growth and Trofimov's Theorem in Section 4.4.

There is a third theorem of Trofimov with a similar feel to it as the two theorems stated above. This theorem involves the concept of an o -automorphisms of a graph. An automorphism g of a connected graph Γ is called an *o -automorphism* if

$$\max\{d(\beta, g\beta) \mid \beta \in V\Gamma, d(\alpha, \beta) \leq n\} = o(n),$$

where α is a fixed vertex. It is easy to show that this property does not depend on the choice of the vertex α . It is also easy to prove that the α -automorphisms form a normal subgroup of $\text{Aut}(\Gamma)$.

Theorem 2.18. ([69, Corollary 1]) *Suppose Γ is a connected locally finite graph and G a group that acts transitively on Γ . Then the following are equivalent:*

- (i) $G \leq o(\text{Aut}(\Gamma))$
- (ii) *There is a G -invariant equivalence relation $\sim$ on the vertex set of Γ such that the equivalence classes of $\sim$ are finite and if K denotes the kernel of the action of G on the equivalence classes then G/K is a finitely generated nilpotent group acting regularly on $\Gamma/\sim$.*

Trofimov's proofs of these three theorems are long and difficult. The proofs mentioned above of the first two theorems, are short, but admittedly, in the proof of Theorem 2.17 the results from the theory of topological groups used are highly non-trivial. It would be interesting to find a topological interpretation of the concept of an α -automorphism. Possibly that could lead to a short proof of Theorem 2.18.

3. The scale function and tidy subgroups

The theory of locally compact groups is the part of the theory of topological groups that has widest appeal and most applications in other branches of mathematics. When looking at locally compact groups there are the connected groups on one end of the spectrum and the totally disconnected groups on the other end.

The fundamental result in the theory of locally compact totally disconnected groups is the theorem of van Dantzig [14] that such a group must always contain a compact open subgroup. A big step towards a general theory was taken in the paper [71] by Willis. The fundamental concepts of Willis's theory are the scale function and tidy subgroups.

Definition 3.1. (TA) Let G be a locally compact totally disconnected group and x an element in G . For a compact open subgroup U in G define

$$U_+ = \bigcap_{i=0}^{\infty} x^i U x^{-i} \quad \text{and} \quad U_- = \bigcap_{i=0}^{\infty} x^{-i} U x^i.$$

Say U is *tidy* for g if

(1) $U = U_+ U_- = U_- U_+$

and

(2) $U_{++} = \bigcup_{i=0}^{\infty} x^i U_+ x^{-i}$ and $U_{--} = \bigcup_{i=0}^{\infty} x^{-i} U_- x^i$ are both closed in G .

(TB) Let G be a locally compact totally disconnected group. The *scale function* on G is defined as

$$s(x) = \min\{|U : U \cap x^{-1} U x| : U \text{ a compact open subgroup of } G\}.$$

The connection between the scale function and tidy subgroups is described in the following theorem due to Willis.

Theorem 3.2. ([77, Theorem 3.1]) *Let G be a locally compact totally disconnected group and $g \in G$. Then $s(g) = |U : U \cap g^{-1} U g|$ if and only if U is tidy for g .*

Now on to something completely different.

3.1. Highly arc transitive digraphs.

Definition 3.3. A directed graph Γ is called *s-arc transitive* if the automorphism group acts transitively on the set of directed paths of length s . If Γ is *s-arc transitive* for all numbers $s \geq 0$ then Γ is said to be *highly arc transitive*.

We also say that a group $G \leq \text{Aut}(\Gamma)$ acts *highly arc transitively* on Γ if G acts transitively on the s -arcs in Γ for all s .

The definition of highly arc transitive digraphs occurs first in the paper [13] by Cameron, Praeger and Wormald. Similar conditions, both for directed and undirected graphs, have been studied by various authors in various context.

Let us start by looking at several examples.

Examples. (i) Let Γ be a directed tree with constant in- and out-valencies. Clearly Γ is highly arc transitive.

(ii) Let Γ be a digraph with the set $\mathbf{Q}$ of rational numbers as a vertex set and (α, β) an edge in Γ if and only if $\alpha > \beta$. Again it is clear that Γ is a highly arc transitive digraph.

(iii) (Cf. [49, Example 1]) Let T_1 denote the regular directed tree in which every vertex has in-valency 1 and out-valency q . Let $L = \dots, \alpha_{-1}, \alpha_0, \alpha_1, \alpha_2, \dots$ be a directed line in T_1 . Define

$$H = \{h \in \text{Aut}(T_1) \mid \text{there is a number } i \text{ such that } h\alpha_i = \alpha_i\}.$$

If $h \in H$ and h fixes some α_i then h also fixes all vertices α_j with $j < i$. One can also see that the orbits of H are infinite and each orbit contains precisely one vertex from L . The orbits of H are called *horocycles*. The horocycles could also be defined in this case without reference to the automorphism group. Then we could define two vertices α and β to be in the same *horocycle* if there is a number n such that the unique path in Γ from α to β starts by going backwards along n arcs and then going forward along n arcs or *vice versa*. Let C_i denote the horocycle containing α_i .

For each $i \in \mathbf{Z}$ take $r - 1$ copies $S_i^1, \dots, S_i^{r-1}$ of T_1 and let $\psi_i^j : S_i^j \rightarrow T_1$ be an isomorphism. The preimage of the horocycle C_i is a horocycle B_i^j in S_i^j . When restricted, ψ_i^j defines an isomorphism between the digraphs spanned by $\text{desc}(B_i^j)$ and $\text{desc}(C_i)$. Use this isomorphism to identify the vertices in $\text{desc}(B_i^j)$ with the vertices in $\text{desc}(C_i)$. Do that for every i and you get a new digraph T_2 . The digraph T_2 is far from being a tree, but if α is a vertex in T_2 then the digraph spanned by $\text{desc}(\alpha)$ is a rooted infinite directed q -ary tree. The vertices in T_2 that did belong to T_1 now all have out-valency equal to q and in-valency equal to r . Look at the part of S_i^j that did not get identified with vertices in T_1 . This part is a union of horocycles, at each horocycle in it we glue $r - 1$ new copies of T_1 in the same fashion. Do this for each i and each horocycle in S_i^j , not belonging to T_1 , and get a digraph T_3 . Continuing in the same fashion we construct a sequence $T_1 \subseteq T_2 \subseteq T_3 \subseteq \dots$ of digraphs. In the end we get a digraph $DL(q, r) = \bigcup T_i$. In this digraph every vertex has in-valency equal to r and out-valency equal to q . If α is a vertex in $DL(q, r)$ then the subdigraph spanned by $\text{desc}(\alpha)$ is an infinite rooted directed q -ary tree and the subgraph spanned by $\text{anc}(\alpha)$ is a rooted tree such that all edges are directed towards the root and the in-valency of every vertex is r and the out-valency is 1. Clearly $DL(q, r)$ is highly arc transitive.

The digraphs $DL(q, r)$ are a directed versions of the Diestel-Leader graphs (defined in [17]) that have been studied by various authors. Woess [81] asked if every locally finite transitive graphs was quasi-isometric to some Cayley graph. It was conjectured by Diestel and Leader that if $q \neq r$ then the graph $DL(q, r)$ is not quasi-isometric to any Cayley graph. This conjecture was proved by Eskin, Fisher and Whyte in [23].

An optimist would hope to find a general classification of locally finite highly arc transitive graphs, but it seems very implausible that any such classifications is possible. But, there is a particular class of highly arc transitive digraphs where one

can give a precise description of their structure. Surprisingly enough this particular class can be used to probe the secrets of Willis's theory.

First we state two simple lemmata from the paper [13] by Cameron, Praeger and Wormald. We proof the second one because it is natural to apply the permutation topology on $\text{Aut}(\Gamma)$ in the proof.

Lemma 3.4. ([13, Proposition 3.10]) *Let Γ be a connected highly arc transitive digraph with finite out-valency. Suppose Γ is not a directed cycle. If α and β are vertices in Γ and there is a directed path of length n from α to β then every directed path from α to β has length n . Furthermore, Γ has no directed cycles.*

Lemma 3.5. *Let Γ be a locally finite highly arc transitive digraph. Take two directed lines $L_1 = \dots, \alpha_{-1}, \alpha_0, \alpha_1, \alpha_2 \dots$ and $L_2 = \dots, \beta_{-1}, \beta_0, \beta_1, \beta_2 \dots$ are directed lines in Γ then there is an automorphism g of Γ such that $g\alpha_i = \beta_i$ for all i .*

Proof. Write $G = \text{Aut}(\Gamma)$ and note that G is locally compact. Using the property that Γ is highly arc transitive we can find an element $g_i \in \text{Aut}(\Gamma)$ such that $g_i\alpha_j = \beta_j$ for all $j \in \{-i, \dots, i\}$. The sequence $(g_i)_{i \in \mathbb{N}}$ is contained in the set $g_1G_{\alpha_0}$ which is compact in the permutation topology on $\text{Aut}(\Gamma)$. Hence this sequence has a convergent subsequence that converges to an element g in $\text{Aut}(\Gamma)$ which has the desired property. $\square$

Proposition 3.6. *Let Γ be a locally finite highly arc transitive digraph and L a directed line in Γ . Then the subdigraph Γ_L spanned by $\text{desc}(L)$ is highly arc transitive and has more than one end.*

Proof. Write $L = \dots, \alpha_{-1}, \alpha_0, \alpha_1, \alpha_2 \dots$. Consider two s -arcs $\beta_0, \dots, \beta_s$ and $\gamma_0, \dots, \gamma_s$ in Γ_L . The vertices β_0 and γ_0 will have a common ancestor α_{i_0} on the line L . Now we can extend the s -arcs to infinite lines $L_\beta = \dots, \beta_{-1}, \beta_0, \beta_1, \beta_2 \dots$ and $L_\gamma = \dots, \gamma_{-1}, \gamma_0, \gamma_1, \gamma_2 \dots$ that both contain the directed ray $\dots, \alpha_{i_0-2}, \alpha_{i_0-1}, \alpha_{i_0}$. Then we can find an element $g \in G$ such that $g\beta_i = \gamma_i$ for all i and because g maps the ray $\dots, \alpha_{i_0-1}, \alpha_{i_0}$ into L we can conclude that $g(\text{desc}(L)) = \text{desc}(L)$, i.e. the subdigraph Γ_L is invariant under g . Whereupon we conclude that Γ_L is highly arc transitive.

Let β' be a vertex in Γ_L . Since $\text{in}_{\Gamma_L}(\beta')$ is finite there clearly is a number i such that $\text{in}_{\Gamma_L}(\beta') \subseteq \text{desc}(\alpha_i)$. Let k be the length of a directed path from α_i to β' (by Lemma 3.4 all directed paths from α_i to β' have the same length). Making use of arc transitivity we conclude that if $\beta \in \text{desc}_k(\alpha_0)$ then there is an element $g \in \text{Aut}(\Gamma_L)$ such that $g(\alpha_i) = \alpha_0$ and $g(\beta') = \beta$. Therefore we see that if (γ, β) is an arc in Γ_L (i.e. $\gamma \in \text{in}_F(\beta)$) then $\gamma \in \text{desc}(\alpha_0)$. More precisely, $\gamma \in \text{desc}_{k-1}(\alpha_0)$. This is so because, if $\alpha_0, \gamma_1, \dots, \gamma_l, \gamma$ is a directed path from α_0 to γ then $\alpha_0, \gamma_1, \dots, \gamma_l, \gamma, \beta$ is a directed path from α_0 to β and thus has length k .

Set $A = \bigcup_{i \geq k} \text{desc}_i(\alpha_0)$ and $A^* = VF \setminus A$. Suppose (γ, β) is an arc from A^* to A . Now $\beta \in \text{desc}_l(\alpha_0)$ for some $l \geq k$. Then $\gamma \in \text{desc}_{l-1}(\alpha_0)$, by the choice of k . Obviously $l = k$ and (γ, β) is an arrow from $\text{desc}_{k-1}(\alpha_0)$ to $\text{desc}_k(\alpha_0)$.

A priori, there is also the possibility that some arc (β, γ) in F goes from A to A^* . But on closer look, this is impossible because then β would be in $\text{desc}_l(\alpha_0)$

for some $l \geq k$ and thus $\gamma \in \text{desc}_{l+1}(\alpha_0) \subseteq A$, and therefore $\gamma \in A$, contradicting the assumption that $\gamma \in A^*$.

The only arcs between A and A^* are going from $\text{desc}_{k-1}(\alpha_0)$ to $\text{desc}_k(\alpha_0)$. The set of such arcs is clearly finite (because $\text{desc}_{k-1}(\alpha_0)$ is finite and the out-valency of vertices in Γ is finite) and by removing them we split the Γ up into components. The two sets $\{\dots, \alpha_0, \alpha_1, \dots, \alpha_{k-1}\}$ and $\{\alpha_k, \alpha_{k+1}, \dots\}$ will belong to different components, so we have at least two infinite components. Hence Γ has more than one end. $\square$

The structure of digraphs like Γ_L in the above proposition is described in the following Theorem.

Theorem 3.7. ([50, Theorem 1]) *Let Γ be a locally finite highly arc transitive digraph. Suppose that there is a line $L = \dots, \alpha_{-1}, \alpha_0, \alpha_1, \dots$ such that $V\Gamma = \text{desc}(L)$.*

Then there exists a surjective homomorphism $\phi : \Gamma \rightarrow T$ where T is a directed tree with in-valency 1 and finite out-valency. The automorphism group of Γ has a natural action on T as a group of automorphisms such that $\phi(g\alpha) = g\phi(\alpha)$ for every $g \in \text{Aut}(\Gamma)$ and every vertex α in Γ . The action of $\text{Aut}(\Gamma)$ on T is highly arc transitive. Furthermore, the fibers $\phi^{-1}(\alpha)$, $\alpha \in VT$, are finite and all have the same number of elements.

Let α be a vertex in a highly arc transitive digraph Γ and denote with c_k the number of vertices in $\text{desc}_k(\alpha)$. Cameron, Praeger and Wormald in [13, Definition 3.5] define the *out-spread* of a vertex in Γ as $\limsup_{k \rightarrow \infty} c_k^{1/k}$. One could also define the *in-spread* of a highly arc transitive digraph in a similar way. Theorem 3.7 implies the following

Theorem 3.8. ([50, Theorem 2]) *The out-spread of a locally finite highly arc transitive digraph is an integer.*

The in-spread can be used to characterize the highly arc transitive digraphs treated in Theorem 3.7.

Theorem 3.9. ([44, Theorem 2.6]) *Let Γ be a locally finite highly arc transitive digraph. The in-spread of Γ is 1 if and only if there is a line L in Γ such that $\text{desc}(L) = V\Gamma$.*

3.2. Tidy subgroups and highly arc transitive digraphs.

Now we turn our attention back to totally disconnected locally compact groups. The following notation will be used extensively in what follows. Let G be a totally disconnected locally compact group and x a fixed element in G . Take a compact open subgroup U . We set $\Omega = G/U$ and let α_0 denote the point in Ω that has U as stabilizer. Then define a digraph $\Gamma = \Gamma_U$ that has Ω as a vertex set and edge set $G(\alpha_0, x\alpha_0)$ – the G -orbit of the ordered pair $(\alpha_0, x\alpha_0)$. Note that Γ need not be connected. For an integer i set $\alpha_i = x^i\alpha_0$. The vertices α_i form a line L in Γ . Observe that x^iUx^{-i} is the stabilizer of α_i in G and U_+ is the stabilizer of the vertices $\alpha_0, \alpha_1, \dots$ and U_- is the stabilizer of the vertices $\alpha_0, \alpha_{-1}, \dots$.

Proposition 3.10. *The subgroup U satisfies condition TA in Definition 3.1 if and only if G acts highly arc transitively on the digraph Γ .*

Proof. Let us start by looking at what happens when the digraph Γ is highly arc transitive. Let $g \in U = G_{\alpha_0}$. Since G is assumed to act highly arc transitively on Γ and G is closed in the permutation topology we deduce from Lemma 3.5 that G acts transitively on the set of lines in Γ . For $i \geq 1$ we set $\beta_i = g\alpha_i$ and let L_1 denote the line $\dots, \alpha_{-1}, \alpha_0, \beta_1, \beta_2, \dots$. We find an element g_- that moves the line L to the line L_1 such that $g_- \alpha_i = \alpha_i$ for $i \leq 0$ and $g_- \alpha_i = \beta_i$ for $i \geq 1$. Note that $g_- \in U_-$. Set $g_+ = g_-^{-1}g$ and note that g_+ fixes all the vertices $\alpha_0, \alpha_1, \dots$ and thus $g_+ \in U_+$. Therefore $g \in U_-U_+$. From this we deduce that U satisfies condition TA.

Assume now that U satisfies condition TA. Take a vertex β in $\text{out}(\alpha_0)$. Then there must be an element $g \in U$ such that $g\alpha_1 = \beta$. Write $g = g_-g_+ \in U_-U_+$ and we see that $g_- \alpha_1 = \beta$. Thus U_- acts transitively on $\text{out}(\alpha_0)$. Now we use induction over s to show that G acts transitively on the set of s -arcs. Suppose we are given an $(s+1)$ -arc $\beta_0, \dots, \beta_s, \beta_{s+1}$. Use the induction hypothesis to find an element $h \in G$ such that $h\alpha_0 = \beta_s, h\alpha_{-1} = \beta_i, \dots, h\alpha_{-s} = \beta_0$. Then by the above, hU_+h^{-1} acts transitively on $\text{out}(h\alpha_0) = \text{out}(\beta_s)$. We pick an element h' from hU_+h^{-1} such that $h'(h\alpha_1) = \beta_{s+1}$. Now we have found an element $h'h$ that moves the $(s+1)$ -arc $\alpha_0, \dots, \alpha_s, \alpha_{s+1}$ to the $(s+1)$ -arc $\beta_0, \dots, \beta_s, \beta_{s+1}$ and can conclude that G acts transitively on the $(s+1)$ -arcs in Γ and also that G acts highly arc transitively on Γ . $\square$

Condition TB can also be translated in to a condition about the graph Γ defined at the start of the section. We use the following lemma.

Lemma 3.11. ([71, Lemma 3]) *Let G be a totally disconnected locally compact group and $x \in G$. Suppose that U is a compact open subgroup of G that satisfies condition TA. Then*

- (a) U_{++} is closed if and only if $U_{++} \cap U = U_+$.
- (b) U_{++} is closed if and only if U_{--} is closed.

In our setting U_{++} is the set of all elements g in G such that there exists a number k such that g fixes $\alpha_k, \alpha_{k+1}, \dots$. The condition that $U_{++} \cap U = U_+$ says that an element in G that fixes α_0 and also $\alpha_k, \alpha_{k+1}, \dots$ for some $k \geq 0$ must also fix $\alpha_1, \dots, \alpha_{k-1}$. If we assume that G acts highly arc transitively on Γ then this implies that $\alpha_0, \dots, \alpha_k$ is the unique directed path in Γ from α_0 to α_k and we conclude that the subgraph spanned by $\text{desc}(\alpha_0)$ is a tree.

On the other hand, if the subgraph spanned by $\text{desc}(\alpha_0)$ is a tree then clearly a group element that fixes α_0 and α_k must fix $\alpha_1, \dots, \alpha_{k-1}$ since these vertices lie on the unique directed path from α_0 to α_k . Hence $U_{++} \cap U = U_+$.

Proposition 3.12. *Suppose U satisfies condition TA. Then U satisfies condition TB if and only if the subgraph spanned by $\text{desc}(\alpha_0)$ is a tree.*

Putting these observation together as a theorem we get.

Theorem 3.13. *Let G be a totally disconnected locally compact group and U a compact open subgroup. Let x be an element in G and define a graph Γ such that the vertex set is G/U and the edge set is $G(\alpha_0, x\alpha_0)$ where α_0 is the vertex in Γ such that $U = G_{\alpha_0}$. Then U is tidy for x if and only if G acts highly arc transitively on Γ and the subgraph spanned by $\text{desc}(\alpha_0)$ is a tree.*

3.3. Using the connection.

In this section G denotes a totally disconnected locally compact group. From the definition of tidy subgroup it is far from obvious that there always is a compact open subgroup of G that is tidy for a given element x in G . Our first task is thus to construct a compact open subgroup U that is tidy for x .

The case where x is periodic (i.e. the subgroup generated by x has compact closure) is easy. If U is a compact open subgroup then we can find a number N such that $x^N \in U$ and the subgroup $U \cap xUx^{-1} \cap \dots \cap x^{N-1}Ux^{-(N-1)}$ is compact and open and normalized by x and thus tidy for x . Hence we will assume in what follows that x is not periodic.

Let V be some compact open subgroup of G . Construct a graph $\Gamma = \Gamma_V$ as done at the start of the last section. From the proof of Proposition 3.10 we see that G acts highly arc transitively on Γ if and only if V_- acts transitively on $\text{out}(\alpha_0)$. Look at the group $V_n = \bigcap_{i=0}^n x^{-i}Vx^i = G_{\alpha_0, \alpha_{-1}, \dots, \alpha_{-n}}$. We claim that there is a number n such that $V_n\alpha_1 = V_-\alpha_1$. Otherwise one could find an element $g_i \in V_i$ for each i such that $g_i\alpha_1 \notin V_-\alpha_1$. The sequence $(g_i)_{i \in \mathbf{N}}$ has a convergent subsequence converging to an element g and clearly this element is in V_- but $g\alpha_1 \notin V_-\alpha_1$ so we have reached a contradiction. Now set $W = V_n$. Note that $W_+ = V_+$. We can use a similar argument as in the first part of Proposition 3.10 to show that W satisfies condition TA. Using this compact open subgroup W to get a compact open subgroup that also satisfies condition TB is more involved and the details will be left out. By finding a compact open subgroup W satisfying TA we have ensured that G acts highly arc transitively on the digraph Γ_W . What is missing is condition TB which would mean that the subgraph spanned by the descendants of a vertex is a tree. Now we use Theorem 3.7 to produce highly arc transitive digraphs where the set of descendants spans a tree and we have proved the following theorem of Willis.

Theorem 3.14. ([71, Theorem 1], see also [49, Theorem 4.1]) *Let G be a totally disconnected locally compact group and x an element of G . Then there is a compact open subgroup U of G that is tidy for x .*

Now we have ensured that there is indeed something to talk about and can start to work our graphs to deduce facts about tidy subgroups and the scale function. First we use Lemma 2.4 to observe the following.

Theorem 3.15. ([71, Corollary 1]) *Let G be a totally disconnected locally compact group. Denote by Δ the modular function on G and by $\mathbf{s}$ the scale function on G . Then, for every $x \in G$,*

$$\Delta(x) = \frac{\mathbf{s}(x)}{\mathbf{s}(x^{-1})}.$$

Proof. Let U_1 and U_2 be compact open subgroups of G such that

$$|U_1 : U_1 \cap x^{-1}U_1x| = \mathbf{s}(x) \quad \text{and} \quad |U_2 : U_2 \cap xU_2x^{-1}| = \mathbf{s}(x^{-1}).$$

Note that

$$|U_1 : U_1 \cap xU_1x^{-1}| \geq \mathbf{s}(x^{-1}) \quad \text{and} \quad |U_2 : U_2 \cap x^{-1}U_2x| \geq \mathbf{s}(x).$$

Now we use Lemma 2.4 and get

$$\frac{\mathbf{s}(x)}{\mathbf{s}(x^{-1})} \geq \frac{|U_1 : U_1 \cap x^{-1}U_1x|}{|U_1 : U_1 \cap xU_1x^{-1}|} = \Delta(x) = \frac{|U_2 : U_2 \cap x^{-1}U_2x|}{|U_2 : U_2 \cap xU_2x^{-1}|} \geq \frac{\mathbf{s}(x)}{\mathbf{s}(x^{-1})}.$$

Hence $\Delta(x) = \mathbf{s}(x)/\mathbf{s}(x^{-1})$. $\square$

This also implies the following corollary.

Corollary 3.16. ([77, Corollary 3.11]) *Let x be an element of a totally disconnected locally compact group G , and U a compact open subgroup of G . Then $|U : U \cap x^{-1}Ux| = \mathbf{s}(x)$ if and only if $|U : U \cap xUx^{-1}| = \mathbf{s}(x^{-1})$.*

Tidy subgroups are related to the scale function as described in Theorem 3.2. The proof of Theorem 3.2 is involved and we will only have a look at the proof that compact open subgroup U such that $\mathbf{s}(x) = |U : U \cap x^{-1}Ux|$ must be tidy. The other part of the proof of Theorem 3.2 showing that if U is tidy then U minimizes $|U : U \cap xUx^{-1}|$ is more tricky and will not be described here.

Consider a compact open subgroup U , a fixed element $x \in G$ and the digraph Γ defined above. By the above, when trying to minimize $|U : U \cap x^{-1}Ux|$ in order to find $\mathbf{s}(x)$, we could equally try to minimize $|U : U \cap xUx^{-1}|$. The latter index is just the out-valency in the digraph Γ_U . When constructing a tidy subgroup for x we start with an arbitrary compact open subgroup V and next find a compact open subgroup V satisfying TA and the out-valency in Γ_V is at most the out-valency of Γ_U . In the second step we ensure that condition TB is satisfied, and in the process the out-valency does not increase. Thus we can be sure that if U minimizes $|U : U \cap xUx^{-1}|$ then U must be tidy for x .

Again we look at a compact open subgroup U and the graph Γ as above. Note that $U\alpha_n = |U : U \cap x^nUx^{-n}|$. If Γ is highly arc transitive then this is precisely the number b_n of vertices β such that there is a directed path of length n from α_0 to β . The out-valency d_+ of Γ is equal to $|U : U \cap xUx^{-1}|$. The subgraph spanned by $\text{desc}(\alpha_0)$ is a tree if and only if $b_n = d_+^n = |U : U \cap xUx^{-1}|^n$. But U is tidy if and only if Γ is highly arc transitive and the subgraph spanned by $\text{desc}(\alpha_0)$ is a tree.

Theorem 3.17. *Let G be a totally disconnected locally compact group and x an element in G . Then a compact open subgroup U is tidy for x if and only if*

$$|U : U \cap x^nUx^{-n}| = |U : U \cap xUx^{-1}|^n$$

for all $n \geq 1$.

Corollary 3.18. *Let G be a totally disconnected locally compact group and x an element in G . Then $\mathbf{s}(x^n) = \mathbf{s}(x)^n$.*

Proof. Let U be a compact open subgroup of G that is tidy for x . It is easy to check that if a subgroup U is tidy for an element x in G then U is also tidy for x^n . Hence

$$\mathbf{s}(x^n) = |U : U \cap x^{-n}Ux^n| = |U : U \cap x^{-1}Ux|^n = \mathbf{s}(x)^n.$$

□

If Γ is highly arc transitive the index $|U : U \cap x^{-n}Ux^n|$ is the number of vertices β such that α_0 is in $\text{desc}_n(\beta)$ and we may think about comparing the scale function and the in-spread of Γ .

Theorem 3.19. ([49, Theorem 7.7]) *Let G be a totally disconnected locally compact group and x an element of G . If V is some compact open subgroup of G then*

$$\mathbf{s}(x) = \lim_{n \rightarrow \infty} |V : V \cap x^{-n}Vx^n|^{1/n}.$$

For a different formulation and a proof see [6, Lemma 4]. This line of thought also gives us information about the case $\mathbf{s}(x) = 1$.

Theorem 3.20. ([49, Corollary 7.8]) *Let G be a totally disconnected locally compact group and x an element of G such that $\mathbf{s}(x) = 1$. If V is some compact open subgroup of G then there is a constant C such that $|V : V \cap x^{-n}Vx^n| \leq C$ for all $n \geq 0$.*

These two results could also be formulated as results about permutation groups.

Theorem 3.21. *Let G be a group acting transitively on a set Ω . Assume that all suborbits of G are finite. Let x be an element in G and α_0 a point in Ω . Set $\alpha_i = x^i\alpha_0$. Then either there is a constant C such that $|G_{\alpha_0}\alpha_n| \leq C$ for all n or the number $|G_{\alpha_0}\alpha_n|$ grows exponentially and $\lim_{n \rightarrow \infty} |G_{\alpha_0}\alpha_n|^{1/n} = C$ for some integer C .*

Remark. Trofimov in [70] defines *generalized x -tracks*. The properties he studies bear similarities to the properties of the directed line $\dots, \alpha_{-1}, \alpha_0, \alpha_1, \alpha_2, \dots$ that are fundamental to the graph theoretical interpretation of Willis's theory in [49]. Trofimov's motivation is entirely different and it is difficult to pinpoint direct similarities between results in [70] and results in [49], but the above Theorem is clearly related to part 3 of [70, Theorem 4.1].

The final illustration of the uses of graphs in Willis's structure theory is a proof of the following theorem.

Theorem 3.22. ([75, Theorem 2]) *Let G be a totally disconnected locally compact group. The set $P(G)$ of periodic elements in G is closed. (An element $x \in G$ is periodic if $\langle x \rangle$ is compact.)*

Proof. The trick is to use the fact that a connected infinite and locally finite highly arc transitive digraph has no directed cycles, see Lemma 3.4.

Suppose x is not periodic, but is in the closure of $P(G)$. Let U be a compact open subgroup of G that is tidy for x . Define the digraph Γ as at the start of Section 3.2. If x is not periodic then the orbit of α_0 under x is infinite and the connected component of Γ that contains α_0 is infinite. (It must contain the line $\dots, \alpha_{-1}, \alpha_0, \alpha_1, \alpha_2, \dots$) The set xU is an open neighbourhood of x and must therefore contain some periodic element g . The fact that $g \in xU = xG_{\alpha_0}$ implies $g\alpha_0 = x\alpha_0 = \alpha_1$. The element g is periodic so the orbit of α_0 under g is finite and therefore there is an integer n such that $g^n(\alpha_0) = \alpha_0$. The sequence $\alpha_0, \alpha_1 = g\alpha_0, \beta_2 = g^2\alpha_0, \dots, \beta_n = g^n\alpha_0 = \alpha_0$ is a directed cycle in Γ . This contradicts Lemma 3.4 mentioned above. Hence we conclude that it is impossible that the closure of $P(G)$ contains any elements that are not periodic. Thus $P(G)$ is closed. $\square$

4. Rough Cayley graphs

Most of the material in this section comes from a paper by Krön and Möller [39]. Let G be a compactly generated totally disconnected locally compact group. In [39] it is shown how one can construct a locally finite connected graph that the group acts transitively on such that the stabilizers of vertices are compact open subgroups of G . The idea is that a rough Cayley graph can be used to study G in much the same way as a ordinary Cayley graph is used when studying a finitely generated group. This idea is demonstrated by looking at the theory of ends of groups and the theory of groups with polynomial growth. In what follows it is described in details how one can construct a rough Cayley graph and that any two such graphs for the same group are quasi-isometric. The applications of this construction to the theory of ends of groups and to groups of polynomial growth are not described in the same details. The work in [39] on ends of groups is based on using tools from the paper [19] by Dunwoody and the book [15] by Dicks and Dunwoody.

4.1. Definition of a rough Cayley graph.

Definition 4.1. ([39, Definition 2.1]) Let G be a topological group. A connected graph Γ is said to be a *rough Cayley graph* of G if G acts transitively on Γ and the stabilizers of vertices are compact open subgroups of G .

Our main task in this section is to show that if G is a compactly generated locally compact group that contains a compact open subgroup then G has a locally finite rough Cayley graph and any two rough Cayley graphs are quasi-isometric to each other. The approach here is different from the approach in [39].

Let G be a compactly generated topological group. For a compact generating set S we form the Cayley graph $\Gamma = \text{Cay}(G, S)$ of G with respect to S . The vertex set of Γ is equal to G and thus carries a topology. The compactness of S and

the continuity of multiplication in G implies that if A is a relatively compact set of vertices in Γ then the neighbourhood of A in Γ is contained in the set $A \cdot S$ and is thus relatively compact also. This was used by Abels [1, 2.3 Heine-Borel-Eigenschaft] to prove that a set A of vertices in Γ is relatively compact if and only if it has finite diameter in the graph metric on Γ .

Definition 4.2. Let G be a group acting transitively on a connected graph Γ . Suppose U is a subgroup of G that contains the stabilizer of some vertex α . The orbit $U\alpha$ is a block of imprimitivity. Let Γ_U denote the quotient graph with respect to the G -congruence whose classes are the translates under G of the set $U\alpha$.

Lemma 4.3. *Let G be a compactly generated topological group. Suppose S is a compact generating set and U is a compact open subgroup of G . Then the graph Γ_U is locally finite.*

Proof. The neighbourhood of a coset gU in Γ is compact and can thus be covered by finitely many cosets of U . Whence Γ_U is locally finite. $\square$

The above gives us the existence of a locally finite rough Cayley graph for a compactly generated group with a compact open subgroup.

Using a result of Sabidussi [58, Theorem 2] we can also see that every locally finite rough Cayley graph for a compactly generated group with a compact open subgroup can be constructed in this fashion. In our terminology the part of Sabidussis Theorem that we will be using reads:

Theorem 4.4. (Cf. [58, Theorem 2]) *Let Γ be a connected graph and G a transitive subgroup of $\text{Aut}(\Gamma)$. Then there is a set S of generators of G such that $\Gamma \approx \text{Cay}(G, S)_U$ where U is the stabilizer in G of some vertex in Γ .*

In our setting we are thinking of a topological group G acting on a locally finite rough Cayley graph Γ , not necessarily faithfully. Take a vertex α in Γ and let U denote the stabilizer in G of α . Looking at Sabidussis proof of his theorem we deduce that if S is the union of U and all elements h in G such that α and $h\alpha$ are adjacent in Γ then $\Gamma \approx \text{Cay}(G, S)_U$. Note also that the orbit $S\alpha$ consists precisely of α and all its neighbours. The graph Γ is by assumption locally finite so S is a finite union of cosets of U and thus compact.

Theorem 4.5. *Let G be a compactly generated group. Any two rough Cayley graphs for G are quasi-isometric.*

The first step in the proof is the following Lemma, whose proof depends on Abels *Heine-Borel eigenschaft* mentioned above.

Lemma 4.6. (i) *Let G be a compactly generated topological group. Suppose S_1 and S_2 are compact generating sets for G . Then the Cayley-graphs $\text{Cay}(G, S_1)$ and $\text{Cay}(G, S_2)$ are quasi-isometric.*

(ii) *Suppose S is a compact generating set and U is a compact subgroup of G . Then the graphs $\text{Cay}(G, S)$ and $\text{Cay}(G, S)_U$ are quasi-isometric.*

Proof. (i) We observe that the elements in S_1 can all be expressed as words of length $\leq C$ in the elements in S_2 for some constant C and *vice versa*. If we have expressed a group element g as a word in the elements in S_1 then we replace each of the elements of S_1 in this word by a word in the elements in S_2 and we may assume that each of these words for the elements in S_2 has length at most C . Now we have expressed g as a word in the elements of S_2 of length at most C times the length of the word in the elements in S_1 . From this we conclude that the two Cayley-graphs $\text{Cay}(G, S_1)$ and $\text{Cay}(G, S_2)$ are quasi-isometric.

(ii) By part (i) we know that $\text{Cay}(G, S)$ and $\text{Cay}(G, S \cup U)$ are quasi-isometric. Thus we may safely assume that S contains U . Each of the right cosets of U has therefore diameter 1 and the quotient graph $\text{Cay}(G, S)_U$ is constructed by contracting each of these cosets. Clearly that operation preserves quasi-isometry. $\square$

Proof. (Theorem 4.5) Suppose Γ_1 and Γ_2 are rough Cayley graphs of G . Then we can find compact generating sets S_1 and S_2 and compact open subgroups U_1 and U_2 such that $\Gamma_1 = \text{Cay}(G, S_1)_{U_1}$ and $\Gamma_2 = \text{Cay}(G, S_2)_{U_2}$. Then Γ_1 is quasi-isometric to $\text{Cay}(G, S_1)$ which in turn is quasi-isometric to $\text{Cay}(G, S_2)$ which is quasi-isometric to $\Gamma_2 = \text{Cay}(G, S_2)_{U_2}$. $\square$

In this context it also worth noting the following which says that the groups that have a locally finite rough Cayley graphs are precisely the topological groups that have a compact open subgroup and are compactly generated.

Proposition 4.7. ([51, Corollary 1]) *Suppose that G is a locally compact topological group and that G acts transitively on a connected locally compact group such that the stabilizers of vertices in G are compact open subgroups of G . Then G is compactly generated.*

The proof of the proposition is based on the Lemma below.

Lemma 4.8. *Let G be a group acting transitively on a locally finite connected graph Γ . Then G has a finitely generated transitive subgroup.*

Proof. Fix a reference vertex α . Denote the neighbours of α by $\beta_1, \dots, \beta_n$. Choose elements $h_1, \dots, h_n$ such that $h_i\alpha = \beta_i$. We claim that $H = \langle h_1, \dots, h_n \rangle$ is transitive on the vertices of Γ . Note that all the vertices in Γ that are adjacent to the vertex α are in the H -orbit of α . Suppose that $\beta = h\alpha$ for some $h \in H$. Then $hh_1h^{-1}\beta, \dots, hh_nh^{-1}\beta$ is an enumeration of all the neighbours of β . Whence the neighbours of β are also contained in the H -orbit of α . Our graph is assumed to be connected so we can conclude that h acts transitively on the vertices. $\square$

Now Proposition 4.7 follows directly because the stabilizer of a vertex plus a finite generating set for H forms a compact generating set for G .

The final step in these basic considerations of rough Cayley graph is the following theorem. The first part is well known.

Theorem 4.9. ([39, Corollary 2.11]) *Let G be a compactly generated topological group that has a compact open subgroup. Assume that H is a cocompact closed subgroup of G . Then H is compactly generated and if Γ_G is a rough Cayley graph for G and Γ_H is a rough Cayley graph for H then Γ_G and Γ_H are quasi-isometric.*

Proof. Let X be a locally finite rough Cayley graph for G . By Lemma 2.3 we know H acts on X with finitely many orbits. Choose a vertex α from Γ . Then there is a number k such that every vertex in Γ is in distance at most k from the orbit $H\alpha$. Now form the graph Γ' which has the same vertex set as Γ but two vertices α, β being adjacent if and only if $d_\Gamma(\alpha, \beta) \leq 2k + 1$. Note that Γ' is also locally finite. Consider the subgraph Δ of Γ' spanned by the vertices in $H\alpha$. Suppose α and β are vertices in Δ and that $\alpha = \alpha_0, \alpha_1, \dots, \alpha_n = \beta$ is a path in Γ (and thus also a path in Γ') from α to β . For each α_i choose a vertex β_i in $H\alpha$ such that $d_\Gamma(\alpha_i, \beta_i) \leq k$. Then $d(\beta_i, \beta_{i+1}) \leq 2k + 1$ so either $\beta_i = \beta_{i+1}$ or β_i and β_{i+1} are adjacent in Γ' . Whereupon we conclude that Δ is connected. Now H acts transitively on a connected locally finite graph with compact open stabilizers of vertices and thus H is compactly generated. From the above it is clear that Δ is quasi-isometric to Γ and the second part of the Theorem follows. $\square$

4.2. Application to FC^- -groups. As mentioned in Section 2.3, an element g of a topological group G is called a FC^- -element if its conjugacy class in G has compact closure. It is an easy exercise to show that the FC^- -elements of G form a normal subgroup $B(G)$ of G . If $G = B(G)$ then G itself is called a FC^- -group. These concepts have been extensively studied, see for example the paper [31] by Grosser and Moskowitz and various papers by Wu and his collaborators, e.g. [83] and [84].

A basic question about the subgroup $B(G)$ is whether it is closed or not. This question was discussed by Tits in [64] where it proved that if G is a connected locally compact group then $B(G)$ is closed. But Tits also gives an example of a locally compact totally disconnected group where $B(G)$ is not closed. Such an example can be described using graphs.

Example. Let Γ denote a directed tree such that each vertex has in-valency 1 and out-valency 2. Choose a directed line $\dots, \alpha_{-1}, \alpha_0, \alpha_1, \alpha_2, \dots$ in Γ . We say that vertices α and β are in the same *horocycle* if there is a number n such that the unique path in Γ from α to β starts by going backwards along n arcs and then going forward along n arcs or *vice versa*. (This concept is also discussed in Section 3.1.) This is an equivalence relation and we let C_i denote the equivalence class of α_i . Define H as the subgroup of $\text{Aut}(\Gamma)$ that stabilizes C_0 . Let G denote the permutation group that H induces on C_0 . We think of C_0 as a metric space, the metric being the restriction of the graph metric on Γ . If g is an element of G that fixes all but finitely many vertices in C_0 then by Lemma 2.16 the element g is an FC^- -element of G . It is also clear that if g and h are elements of G , both with finite support, then there is a conjugate of h whose support is disjoint from the support of g . Let g_i be an element of G with finite support such that there is a

vertex $\alpha \in C_0$ such that $d(\alpha, g_i\alpha) = 2i$. (We could define g_i explicitly by defining β_{i+1} as the vertex such that $(\alpha_{-i-2}, \beta_{i+1})$ is an arc in Γ and $\beta_{i+1} \neq \alpha_{-i-1}$ and then let g transpose the two outward directed arcs from β_{i+1} .) We may assume that $\text{supp } g_i \cap \text{supp } g_j = \emptyset$ if $i \neq j$. All the g_i 's are contained in $B(G)$ and $h_i = g_1 \dots g_i$ is also in $B(G)$. Then we define g as the limit of the sequence of the h_i 's ($g\alpha = h_i\alpha$ if α is in $\text{supp } h_i$ and $g\alpha = \alpha$ if α is not in the support of any of the h_i 's). Clearly g is not a bounded isometry of C_0 and thus g is not in $B(G)$. Hence $B(G)$ is not closed.

The construction of a rough Cayley graph can be used in the study of FC^- -elements in groups. Trofimov in [68] proved the following.

Theorem 4.10. ([68], cf. [81, Theorem 3]) *Let Γ be a vertex transitive connected locally finite graph. The subgroup of bounded automorphisms in $\text{Aut}(\Gamma)$ is closed in $\text{Aut}(\Gamma)$.*

This theorem plus the construction of a rough Cayley graph gives the following.

Theorem 4.11. ([51, Theorem 2]) *Let G be a compactly generated totally disconnected locally compact group. Then the subgroup of FC^- -elements is closed in G .*

The proof is simple. One constructs a locally finite rough Cayley graph for G . The action of G on Γ gives a continuous homomorphism $G \rightarrow \text{Aut}(\Gamma)$. It is easy to show that the kernel of this homomorphism is compact and the image is closed in $\text{Aut}(\Gamma)$ (see [51, Corollary 1]). The subgroup of bounded automorphism in $\text{Aut}(\Gamma)$ is closed by Trofimov's theorem above and the subgroup of FC^- -elements is closed in G because it is precisely the preimage of the subgroup of bounded automorphisms in $\text{Aut}(\Gamma)$.

4.3. Rough ends. The ends of a graph can be thought of as representing the “different” ways of going to infinity, or as forming a boundary for the graph.

Recall that a *ray* in a graph is a sequence $\alpha_0, \alpha_1, \dots$ of distinct vertices such that α_i is adjacent to α_{i+1} for all i . The graph theoretic approach is to define the ends of a graph Γ as equivalence classes of rays. Two rays are equivalent (i.e. are in the same *end*) if there is the third ray that intersects both infinitely often. This definition goes back to the paper [32] by Halin.

Ends can also be defined with reference to connected components when finite sets of edges are removed from the graph. For a finite set Φ of edges define $\mathcal{C}_\Phi$ as the set of connected components of $\Gamma \setminus \Phi$. Suppose Φ_1 and Φ_2 are two finite sets of edges such that $\Phi_1 \subseteq \Phi_2$. There is a natural map $\mathcal{C}_{\Phi_2} \rightarrow \mathcal{C}_{\Phi_1}$ that takes a component c of $\Gamma \setminus \Phi_2$ to the component of $\Gamma \setminus \Phi_1$ that contains c . With these maps the collection of all the sets $\mathcal{C}_\Phi$ where Φ ranges over all finite sets of edges in Γ becomes an inverse system. The space of ends $\Omega\Gamma$ (with the natural topology of the inverse limit) of Γ is then defined as the inverse limit of this system. One could also look at the components of $\Gamma \setminus \Phi$ where Φ is a finite set of vertices, but when the graph is locally finite the inverse limits are homeomorphic. This approach goes

back even further to the thesis of Freudenthal in 1931. In later works Freudenthal and Hopf built up a theory of ends of spaces, see [28], [29] and [33].

The *co-boundary* of a set $c \subseteq V\Gamma$ is the set of all edges in Γ such that one of its end vertices is on c and the other is in $V\Gamma \setminus c$. Denote the co-boundary of c with δc . Suppose $|\delta c| < \infty$. From the graph theoretical definition of an end of a graph one concludes that if c contains all but finitely many vertices from a ray R then c also contains also all but finitely many vertices from any ray in the same end as R . Thus we can say that the end belongs to c . Define Ωc as the set of ends that belong to c . The topology on the space of ends has as a basis the sets Ωc where $c \subset V\Gamma$ and $|\delta c| < \infty$. The graph $\Gamma \setminus \delta c$ is not connected. If ω_1 and ω_2 are two ends of Γ and ω_1 belongs to c and ω_2 belongs to $V\Gamma \setminus c$ then we can say that δc separates ω_1 and ω_2 .

It is easy to show that a quasi-isometry between two locally finite connected graph graphs induces a homeomorphism between the end spaces of the graphs. In particular the number of ends is a quasi-isometry invariant.

For a detailed introduction to ends of graphs the reader can consult [47] or [16].

4.3.1. Stallings theorem. For a finitely generated group the number of ends is defined as the number of ends of a Cayley graph of the group with respect to a finite generating set. The number of ends does not depend on the choice of generators when constructing the Cayley graph, because as pointed out above the number of ends is a quasi-isometry invariant. In group theory it mainly the number of ends that is of interest. The number of ends of a finitely generated group is either 0, 1, 2 or ∞ .

Looking at compactly generated totally disconnected locally compact group G and a rough Cayley graph Γ of G it is natural to define the *space of rough ends* of G as the space of ends Γ . Since any two rough Cayley graphs of G are quasi-isometric then the choice of Γ is immaterial. The number of rough ends of G can be either 0, 1, 2 or ∞ .

A finitely generated group with 0 ends is finite, but a compactly generated totally disconnected locally compact group with 0 rough ends is compact.

Finitely generated groups with precisely two ends are characterized by the following result which is a conjunction of results of Hopf and C. T. C. Wall.

Theorem 4.12. ([33, Satz 5] and [72, Lemma 4.1]) *Let G be a finitely generated group. Then the following are equivalent: Let G be a finitely generated group. Then the following are equivalent:*

- (i) G has precisely two ends;
- (ii) G has an infinite cyclic subgroup of finite index;
- (iii) G has a finite normal subgroup N such that G/N is either isomorphic to the infinite cyclic group or to the infinite dihedral group.

For compactly generated totally disconnected locally compact group we have the following analogue.

Theorem 4.13. (Cf. Miller and Seifter, 1998) *Let G be a compactly generated totally disconnected locally compact group. Suppose that the space of rough ends*

has precisely two points. Then G has a compact open normal subgroup N such that G/N is either isomorphic to $\mathbf{Z}$ or D_∞ .

Finitely generated groups with infinitely many ends are characterized by a well known result of Stallings from 1968.

Theorem 4.14. [63] *Suppose G is a finitely generated group with more than one end. Then G can be written as a non-trivial free product with amalgamation $B *_C D$ where C is finite, or G can be written as a non-trivial HNN-extension $B *_C x$ where C is finite.*

In 1974 Abels [1] proved an analogue of Stallings theorem for topological groups. Abels uses similar ideas in his proof as used by Stallings. In [39] essentially the same result as Abels proved is proved by using Dunwoodys theory of structure trees (see [19] and [15]) and the Bass-Serre theory of group actions on trees.

Theorem 4.15. (Cf. Abels, 1974) *Let G be a compactly generated totally disconnected locally compact group. Suppose that some (equivalently, any) rough Cayley graph G has infinitely many ends. Then the group G splits over some compact open subgroup, more precisely, $G = A *_C B$ or $G = A *_C x$ where A and B are closed open compactly generated subgroups of G and C is a compact open subgroup.*

That it is possible to write G as either a free product with amalgamation $G = A *_C B$ or an HNN-extension $G = A *_C x$ is often expressed by saying that G splits over C . Using this expression Stallings theorem becomes the statement that a finitely generated group with infinitely many ends splits over a finite subgroup, and Theorem 4.15 above becomes the statement that a compactly generated totally disconnected locally compact group with more than one end splits over a compact open subgroup. Theorem 4.9 says that if H is a closed cocompact subgroup of G then the rough Cayley graphs of H and G are quasi-isometric, and thus have the same number of ends. The above Theorem hence leads to the following corollary.

Corollary 4.16. ([39, Corollary 3.22]) *Let G be a totally disconnected locally compact group and H a closed cocompact subgroup. Then G splits over a compact open subgroup if and only if H splits over a compact open subgroup.*

4.3.2. Free subgroups. A well known theorem of Gromov (also proved independently by Woess [80]) says that a finitely generated group is quasi-isometric to a tree if and only if it has a finitely generated free subgroup of finite index.

The analogue of this for compactly generated totally disconnected group is the following.

Theorem 4.17. ([39, Theorem 3.28]) *Let G be a compactly generated totally disconnected locally compact group.*

(i) *Some (hence, every) rough Cayley graph of G is quasi-isometric to a tree if and only if G has an expression as a fundamental group of a finite graph of groups such that all the vertex and edge groups are compact open subgroups of G .*

(ii) *Assume also that the group G is unimodular. Then some (hence, every) rough Cayley graph of G is quasi-isometric to some tree if and only if G has a finitely generated free subgroup that is cocompact and discrete.*

The proof of this consists of piecing together various bit of information about ends, quasi-isometries and structure trees from various sources (see [80], [38], [40], [46] and [65]). The essential result used about graphs quasi-isometric to trees is that a transitive locally finite graph is quasi-isometric to a tree if and only if it has no *thick ends* (cf. [40, Theorem 5.5]). This property can then be used to show that for a locally finite transitive graph that is quasi-isometric to a tree one can find a locally finite structure tree on which the automorphism group of the original graph acts.

In the second part the existence of a cocompact discrete finitely generated free subgroup follows from the theory of tree lattices, see [3].

Corollary 4.18. *Let G be a totally disconnected locally compact group. If G has a cocompact finitely generated free discrete subgroup then G splits over some compact open subgroup and G can be written as $G = A *_C B$ or $G = A *_C x$ where A, B and C are compact open subgroups of G .*

The above Corollary implies a special case of a result of Mosher, Sageev and Whyte [53, Theorem 9].

Corollary 4.19. *Let G be a totally disconnected locally compact group. If G has a cocompact finitely generated free discrete subgroup then G has an action on a locally finite tree such that G fixes neither an edge nor a vertex.*

Consider a finitely generated group H and a Cayley graph Γ of H . The action of H on the Cayley graph gives an embedding of H as a closed cocompact subgroup into the totally disconnected locally compact group $G = \text{Aut}(\Gamma)$. Willis in [78, Section 6] asks about if one can just from the fact that G is the automorphism group of some Cayley graph of H place limits on various invariants of G . For example he asks if it is possible to deduce that there are only finitely many prime numbers that occur as factors in $\mathbf{s}(x)$ for $x \in G$ (the scale function $\mathbf{s}$ is discussed in Section 3. This question is motivated by the following result.

Theorem 4.20. ([76, Theorem 3.4]) *Let G be a compactly generated totally disconnected locally compact group. Then there are finitely many primes $p_1, p_2, \dots, p_t$ the number $\mathbf{s}_G(x)$ for all elements $x \in G$ can be written in the form $\mathbf{s}_G(x) = p_1^{s_1} p_2^{s_2} \cdots p_t^{s_t}$.*

Baumgartner [5] has applied the result of Mosher, Sageev and Whyte to the program suggested by Willis. Baumgartner has also extended the scope of the program by considering not only the special type of embedding $G \rightarrow \text{Aut}(\Gamma)$ where Γ is a Cayley graph of H . A topological group G is called an *envelope* of a group H if H embeds as a closed cocompact subgroup of G .

Theorem 4.21. [5, Corollary 11]) *Let H be a virtually free group of rank at least 2. Then there are finitely many primes $p_1, p_2, \dots, p_t$ the number $\mathbf{s}_G(x)$ for all totally disconnected locally compact envelopes G of H all elements $x \in G$ can be written in the form $\mathbf{s}_G(x) = p_1^{s_1} p_2^{s_2} \cdots p_t^{s_t}$.*

The following result comes from a totally different direction but also indicates the possible fruitfulness of the study of envelopes and embeddings of a finitely generated group into automorphism groups of its Cayley graphs.

Theorem 4.22. [52, Theorem 4.1]) *Let N be a finitely generated torsion free nilpotent group and Γ a Cayley graph of N with respect to some finite generating set. Set $G = \text{Aut}(\Gamma)$. Then G is discrete in the permutation topology and N embeds into G as normal subgroup.*

This theorem is proved with the aid of Theorem 2.18.

4.3.3. Accessibility.

Definition 4.23. A finitely generated group is said to be *accessible* if it has an action on a tree T such that:

- (i) the number of orbits of G on the edges of T is finite;
- (ii) the stabilizers of edges in T are finite subgroups of G ;
- (iii) every stabilizer of a vertex in T is a finitely generated subgroup of G and has at most one end.

The question of C. T. C. Wall, [73], of whether or not every finitely generated group is accessible was open for a long time and served as a motivation for several important developments in combinatorial group theory. Amongst those developments arising from the accessibility problem is Dunwoody's theory of structure trees. In the end it turns out that there are finitely generated groups that are not accessible, the first examples constructed by Dunwoody [21].

Definition 4.24. ([65, p. 249]) Let Γ be a connected locally finite graph. If there is a number k such that any two distinct ends can be separated by removing k or fewer vertices from Γ then the graph Γ is said to be *accessible*.

Thomassen and Woess [65, Theorem 1.1] show that a finitely generated group is accessible if and only if every Cayley graph with respect to a finite generating set is accessible.

The definition of accessibility can be rephrased to fit into our framework of compactly generated totally disconnected locally compact groups.

Definition 4.25. A compactly generated totally disconnected locally compact group is said to be *accessible* if it has an action on a tree T such that:

- (i) the number of orbits of G on the edges of T is finite;
- (ii) the stabilizers of edges in T are compact open subgroups of G ;
- (iii) every stabilizer of a vertex in T is a compactly generated subgroup of G and its rough Cayley graphs have at most one end.

Accessibility of compactly generated totally disconnected locally compact groups can then be linked to rough Cayley graphs in analogous way as Thomassen and Woess link accessibility of finitely generated groups to Cayley graphs.

Theorem 4.26. ([39, Theorem 3.27]) *Let G be a compactly generated totally disconnected locally compact group. Then G is accessible if and only if every rough Cayley graph of G is accessible.*

It is pointed out in [56, Theorem 0.4] the property of a graph being accessible is invariant under quasi-isometries. Thus, if G has a cocompact closed accessible subgroup then G itself must be accessible. Dunwoody showed [20] that a finitely presented group is accessible. Combining Dunwoody Theorem with the observation above we get the following Corollary.

Theorem 4.27. *Let G be a compactly generated totally disconnected locally compact group. If G has a cocompact finitely presented subgroup then G is accessible.*

4.3.4. Ends of pairs of groups. By Stallings Theorem the ends of a finitely generated group can be used to detect if the group splits over a finite subgroup. The concept of ends of pairs of groups is an attempt to define a geometric invariant that can be used to detect splittings of G over subgroups that are not necessarily finite. This concept was first introduced in the papers by Houghton [34] and Scott [61]. For a survey of these and related concepts see [74].

Definition 4.28. (Cf. [61, Lemma 1.1]) Let G be a finitely generated group and C a subgroup of G . Let Γ be a Cayley-graph of G with respect to some finite generating set. The number of ends of the pair G and C is defined as the number of ends of the quotient graph Γ/C . (Quotient with respect to the orbits of C on Γ .)

Of-course one must show that this concept is not dependent on the choice of generators used to construct Γ . Houghton in [34] discusses ends of compactly generated Hausdorff groups and builds up his theory in an algebraic fashion.

The problem with the definition of ends of pairs of groups is that the group G does not act on the graph Γ/C . As mentioned above a transitive locally finite graph always has 0, 1, 2 or infinitely many ends, a pair of groups can have any number of ends, see [61, Example 2.1].

Kropholler conjectures the following:

Conjecture 4.29. *Let G be a finitely generated group and H a subgroup of G . If $e(G, H) \geq 2$ then G splits over a subgroup that is commensurable with a subgroup of H .*

Various results in this vein under additional hypothesis are known. One of the more simple ones involves *commensurability*. Two subgroups U_1 and U_2 of a group G are said to be commensurable if their intersection $U_1 \cap U_2$ has finite index both in U_1 and U_2 . Furthermore the *commensurator* of $U \leq G$ in G is defined as the subgroup of all elements $g \in G$ such that U and gUg^{-1} are commensurable.

Theorem 4.30. ([22, p. 30]) *Let G be a finitely generated group and C a finitely generated subgroup of G . If $e(G, C) > 1$ and the commensurator of C is the whole group G then G splits over a subgroup commensurable with C .*

This result has also been proved in papers by Niblo [55, cf. Theorem B] and Scott and Swarup [62, Theorem 3.12].

Looking at the totally disconnected locally compact groups then we notice that two compact open subgroups are always commensurable and that the commensurator of a compact open subgroup is the whole group G .

Consider a totally disconnected locally compact group G and U a subgroup such that the commensurator of U is the whole of G . Let Γ be a Cayley graph of G with respect to some compact set of generators. The graph X_U is a rough Cayley graph of G but X/U is the quotient graph used above to define the ends of the pair of groups G and U . Under these assumptions it can be shown that both these graphs are locally finite and the graph X_U has at least as many ends as the graph X/U (see [39, Theorem 3.57]). In [39, Section 3.7] there is further discussion of how the concepts of ends of pairs of groups, coends and rough ends relate and how these concepts can be interpreted graph theoretically. Amongst other things these considerations lead to a prove of Theorem 4.30 above.

4.4. Polynomial growth. Recall from Section 2.3 that a connected locally finite graph is said to have *polynomial growth* if for every vertex the number of vertices in distance less than or equal to n grows polynomially with n . A finitely generated group is said to have polynomial growth if some (hence, every) Cayley graph with respect to a finite generating set had polynomial growth.

There is a natural definition of the concept of polynomial growth for compactly generated locally compact groups.

Definition 4.31. Let G be a locally compact group generated by a compact symmetric neighbourhood of the identity V . Set $V^n = \{g_1 g_2 \cdots g_n \mid g_i \in V\}$. Let μ denote a Haar measure on G . If there are constants c and d such that $\mu(V^n) \leq cn^d$ for all natural numbers n then we say that G has polynomial growth.

One can now relate the property that a compactly generated totally disconnected locally compact group has polynomial growth to rough Cayley graphs.

Theorem 4.32. ([39, Theorem 4.4]) *Let G be a compactly generated totally disconnected locally compact group and Γ some rough Cayley graph for G . Then Γ has polynomial growth if and only if G has polynomial growth (in the sense of Definition 4.31).*

Viewed in this context Trofimovs theorem about automorphism groups of graphs with polynomial growth, Theorem 2.17, can now be seen as a version of Gromovs Theorem for compactly generated totally disconnected locally compact groups. Losert in [41] proved a more general version for topological groups and Woess in [81] used Loserts result to prove Trofimovs theorem. The following theorem [39, Theorem 4.6] is a combination of Theorem 4.32 and Trofimovs theorem, but can also be seen as Corollary to Losert results.

Theorem 4.33. (Cf. Trofimov [67] and Losert [41]) *Let G be a compactly generated totally disconnected locally compact group. Then G has polynomial growth if and only if G has a normal compact open subgroup K such that G/K is a finitely generated almost nilpotent group.*

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Reykjavk, Iceland
E-mail: roggi@raunvis.hi.is